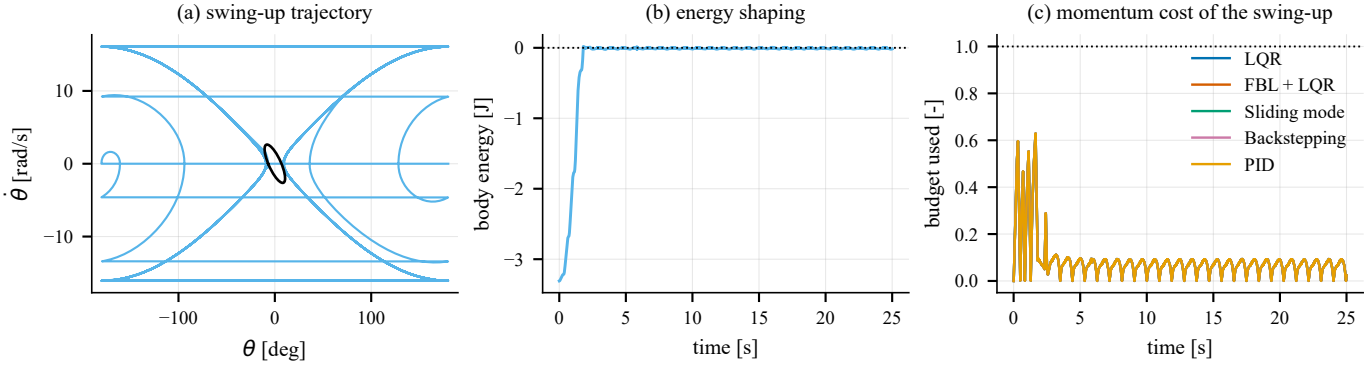

(a) swing-up trajectory
(b) energy shaping
(c) momentum cost of the swing-up
$\dot{\theta}$ [rad/s]
θ [deg]
body energy [J]
time [s]
budget used [-]
time [s]
LQR
FBL + LQR
Sliding mode
Backstepping
PID